

Some Simple facts:Fact 1: If μ_1, μ_2 2 measures on $(\Omega, \mathcal{F})$ Then $\mu_3 = \alpha\mu_1 + \beta\mu_2$, $\alpha, \beta \geq 0$ is a meas.

and $\int g d\mu_3 = \alpha \int g d\mu_1 + \beta \int g d\mu_2$

$$\int f^2 d\mu_1 \leq \int f^2 d\mu_1 + \int f^2 d\mu_2 = \int f^2 d\mu.$$

Fact 2: If $B \in \mathcal{F}$ is fixed, μ is a meas on $(\Omega, \mathcal{F})$ $\nu(A) = \mu(A \cap B)$, $A \in \mathcal{F}$ is a meas.Fact 3: If $f \geq 0$ a.e. μ , Then $\int f d\mu \geq 0$.But if $\int f d\mu \geq 0$, we may not have $f \geq 0$ a.e. μ However if $\int_A f d\mu \geq 0$ for all $A \in \mathcal{F}$, Then $f \geq 0$ a.e. μ Pf: enough to show $\mu(f < -\frac{1}{n}) = 0 \quad \forall n \geq 1$. (check)if not, $\exists n_0 \geq 1$ s.t. $\mu(\underbrace{f < -\frac{1}{n_0}}_A) > 0$.

$$\int_A f d\mu = \int_{\{f < -\frac{1}{n_0}\}} f d\mu < -\frac{1}{n_0} \cdot \underbrace{\mu(f < -\frac{1}{n_0})}_{\neq 0} < 0. \quad \text{contradiction.}$$

Fact 4: Similar to fact 3, one can show

if $\int_A f d\mu = \int_A g d\mu \quad \forall A \in \mathcal{F}$

then $f = g$ a.e. μ Pf: start by $A = \{f < g\}$, $\int_A (g - f) d\mu = 0$.
(check)ex: 4.1.2' : $\Omega = \mathbb{Z}$, ν = counting meas.

$$\mu(A) = \sum_{i \in A \cap \mathbb{Z}_+} \frac{e^{-\lambda} \lambda^i}{i!} \leftarrow \text{Poisson}(\lambda)$$

 $\mu \ll \nu$, but $\nu \not\ll \mu$.eg: take $A = \{-2, -1\}$, $\mu(A) = 0$ but $\nu(A) = 2$ ex: 4.1.2'' $\Omega = \mathbb{Z}$, $\nu(A) = \#(A \cap \mathbb{Z}_-)$
 $\uparrow$
 $\{-1, -2, \dots\}$

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 μ as above then $\uparrow$ counting meas. $\{-1, -2, \dots\}$.

$$\mu \perp \nu$$

take $B = \mathbb{Z}_-$, then $\mu(B) = 0 = \nu(B^c)$.

$$\int_{A \cap B} f d\mu = \int 1_{A \cap B} f d\mu = \int 1_A 1_B f d\mu = \int_A 1_B f d\mu \text{ etc.}$$

Some "fun" RN calculations:

$$\mu(A) = \int_A \phi(x) d\mu(x).$$

① $\mu = \text{Normal}(0,1)$, $\nu = m (= \text{leb})$.

if ν is ref. meas., then $\mu \ll \nu$ and $\frac{d\mu}{d\nu} = \phi$ a.e.(m)

So Leb. decomp of μ w.r.t ν (ref) is: $\mu = \mu + 0$
 $\ll \nu \perp \nu$.

if μ is the ref. meas.

here $m \ll \mu$ and $\frac{dm}{d\mu} = \frac{1}{\phi}$ (prop 4.1.2 (iii)) ✓
 since $\frac{\phi}{\text{a.e.}(m)} \geq 0$

note: $\mu(A) = \int_A \phi dm$

$$m(A) = \int 1_A dm = \int 1_A \cdot \frac{1}{\phi} \cdot \phi dm \leftarrow$$

$$= \int 1_A \frac{1}{\phi} d\mu = \int_A \frac{1}{\phi} d\mu.$$

② μ is discrete; p_i prob. on $x_i \in \mathcal{I}$ (ctble) eg $\mathcal{I} = \mathbb{Z}$

$$\mu(A) = \sum_{x_i \in \mathcal{I}} p_i 1_A(x_i) = \int_A p(x) d\nu(x) \text{ (check)}$$

$$\text{where } \nu(A) = \#(A \cap \mathcal{I}).$$

$$p(x) = \begin{cases} p_i, & x = x_i, i \geq 1 \\ 0, & x \notin \mathcal{I}. \end{cases}$$

Hence $\mu \ll \nu$ and $\frac{d\mu}{d\nu} = p$ a.e.(ν).

So, if μ is $\text{Bin}(n, p)$ then with $\nu = \#(A \cap \mathbb{Z})$

$$\frac{d\mu}{d\nu} = \text{Bin}(n, p) \text{ pmf.}$$

$$\text{or } \left(\frac{d\mu}{d\nu} \right)(x) = \binom{n}{x} p^x (1-p)^{n-x}, \text{ if } x \in \{0, \dots, n\} \text{ } \left. \vphantom{\frac{d\mu}{d\nu}} \right\} x \in \mathbb{R}$$

$$\text{or } \left(\frac{d\mu}{d\nu}\right)(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & \text{if } x \in \{0, 1, \dots, n\} \\ 0, & \text{if } x \notin J \end{cases} \quad x \in \mathbb{R}$$

$\{0, 1, \dots, n\}$

③ $\mu = \text{Bin}(n, p)$, $\nu = \text{Poi}(\lambda)$, $\nu_+(A) = \#(A \cap \mathbb{Z}_+)$
 larger support than μ .

$$\frac{d\mu}{d\nu_+} = f_\mu \quad (= \text{Bin pmf function as above})$$

easy to see $\mu \ll \nu$.

$$\frac{d\nu}{d\nu_+} = f_\nu \quad (= \text{Poi pmf})$$

$$> 0 \quad \text{a.e. } (\nu_+) \rightarrow \text{check.}$$

$$\text{so } \frac{d\nu_+}{d\nu} = \frac{1}{f_\nu} \Rightarrow \frac{d\mu}{d\nu} = \frac{d\mu}{d\nu_+} \cdot \frac{d\nu_+}{d\nu} = \frac{f_\mu}{f_\nu}$$

If $\mu = \text{Bin}$ is ref meas. Then ν is not dom by μ .

So with μ as ref meas,

$$\left. \begin{aligned} \nu_a(A) &= \nu(A \cap \{0, 1, \dots, n\}) \\ \nu_s(A) &= \nu(A \cap \{n+1, \dots\}) \end{aligned} \right\} \nu = \nu_a + \nu_s$$

$$\text{and } \frac{d\nu_a}{d\mu} = \frac{f_{\text{poi}}}{f_{\text{bin}}} \mathbb{I}_{\{0, 1, \dots, n\}} \rightarrow \text{check.}$$

④ $\mu = \chi^2(n)$, $\nu = \text{Cauchy}(0, 1)$. next.
 support: $(0, \infty)$ larger support.
 support $(-\infty, \infty)$

$$\mu(A) = \int_A f_\mu d\mu$$

$$\nu(A) = \int_A f_\nu d\mu$$

$$\rightarrow f_\nu(x) = \begin{cases} g(x), & x > 0 \\ 0, & \text{o.w.} \end{cases}$$

$$\text{where } g(x) = \frac{e^{-x/2} x^{n/2-1}}{2^{n/2} \Gamma(n/2)}, x > 0$$

$$f_\nu(x) = \frac{1}{\pi(1+x^2)}, -\infty < x < \infty$$

$$> 0 \quad \forall x \in \mathbb{R}.$$

Clearly (check)

$$\mu \ll \nu$$

$$\text{and } \frac{d\mu}{d\nu} = \frac{d\mu}{d\mu} \cdot \left(\frac{d\mu}{d\nu}\right) \geq 0.$$

$$= f_\mu \cdot \frac{1}{f_\nu} = \frac{f_\mu}{f_\nu}.$$

1010 ν is not measure $\nu \not\ll \mu$.

$$= \int f v \, d\mu - \int f v \, d\mu$$

alt'ly, if μ is ref measure $\nu \not\ll \mu$.

$$A = (-\infty, 0), \mu(A) = 0 \text{ but } \nu(A) \neq 0$$

$$\text{So, we take } \nu_a(A) = \nu(A \cap (0, \infty))$$

$$\nu_s(A) = \nu(A \cap (-\infty, 0])$$

to find $\frac{d\nu_a}{d\mu}$:

$$\text{note } \underline{\nu_a(A)} = \nu(A \cap (0, \infty))$$

$$= \int_A f v \, 1_{(0, \infty)} \, d\mu = \int_A f v \, d\mu'$$

$$\Rightarrow \frac{d\nu_a}{d\mu} = f v \text{ a.e. } (\mu')$$

$$\text{where } \mu'(A) = \mu(A \cap (0, \infty))$$

if we define
note $\mu'(A) = \mu(A \cap (0, \infty))$

$$\text{i.e. } \int \underline{1_A} \, d\mu' = \int \underline{1_A} \, 1_{(0, \infty)} \, d\mu$$

$$\text{so } \int h \, d\mu' = \int h \, 1_{(0, \infty)} \, d\mu$$

Pf

$$\text{Note } \mu(A) = \int_A f \mu \, d\mu = \int g \cdot 1_{(0, \infty)} \, d\mu = \int g \, d\mu'$$

$$\text{and } g > 0 \text{ a.e. } \mu'$$

$$\text{or, } f \mu > 0 \text{ a.e. } \mu' \text{ (not } \mu)$$

$$\text{So } \frac{d\mu}{d\mu'} = g \text{ (or } f \mu) > 0 \text{ a.e. } (\underline{\underline{\mu'}})$$

$$\text{So } \frac{d\nu_a}{d\mu} = \frac{d\nu_a}{d\mu'} \cdot \frac{d\mu'}{d\mu} = \frac{f v}{g} \text{ a.e. } \mu' \text{ (also a.e. } (\mu))$$

$$\text{So } \left(\frac{d\nu_a}{d\mu} \right)(x) = \frac{1}{\pi(1+x^2)} \text{ a.e. } (\mu')$$

$$\text{So } \left(\frac{d\nu_a}{d\mu} \right)(x) = \left(\frac{\frac{1}{\pi(1+x^2)}}{\frac{e^{-x/2} x^{n/2-1}}{2^{n/2} \Gamma(n/2)}} \right) \quad \text{a.e. } (m'.)$$

can also take $\left(\frac{d\nu_a}{d\mu} \right)(x) = \left(\quad \right) \mathbb{I}_{(0,\infty)}$

$$\textcircled{5} \quad \mu = \overset{\mu_1}{\chi^2_{(2)}} + \text{Bin}\left(3, \frac{1}{4}\right) \quad , \quad \textcircled{\nu} \overset{\text{ref.}}{=} \text{Cauchy}(0,1)$$

$\downarrow \qquad \qquad \downarrow$
 $\mu_a \ll \nu \quad \quad \mu_s \perp \nu$

$$\textcircled{6} \quad \mu = \overset{\mu_1}{\text{Cauchy}(0,1)} + \overset{\mu_2}{\text{Bin}\left(3, \frac{1}{4}\right)} \quad , \quad \textcircled{\nu} \overset{\text{ref.}}{=} \chi^2_{(n)}$$

$\swarrow \qquad \qquad \underbrace{\hspace{2cm}}$
 $\mu_a(A) = \mu_1((0, \infty) \cap A) \quad , \quad \mu_s^{(A)} = \mu_1((-\infty, 0] \cap A) + \mu_2.$

$$\textcircled{7} \quad \mu = \text{Cauchy}(0,1) + \text{Bin}\left(3, \frac{1}{4}\right) \quad , \quad \textcircled{\nu} \overset{\text{ref.}}{=} \text{Poisson}(8)$$

$\downarrow \qquad \qquad \downarrow$
 $\mu_s \quad \quad \mu_a.$